

ATVB IN FOCUS:

Sleep and Cardiovascular Disease

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Sleep and Risk for Heart Disease and Stroke

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ABSTRACT: Getting healthy sleep was recognized in 2022 by the American Heart Association as a key health behavior of Life's Essential 8 based on growing evidence of its impact on cardiovascular health. American Heart Association guidelines recommend that adults aged ≥ 20 years get on average 7 to 9 hours of sleep per night, based on self-reported measures. However, many adults report getting inadequate sleep duration, a trend expected to worsen through 2050. For these reasons, the relationship between sleep, including its multidimensional components (eg, timing, efficiency, regularity, and architecture), and cardiovascular disease must be evaluated further. In this review, we summarize the current evidence on the association of multidimensional sleep with heart disease and stroke risk. In addition, we discuss the advantages and limitations of various sleep assessments from self-report to direct measurement via novel digital health technologies.

Key Words: atrial fibrillation ■ cardiovascular diseases ■ heart disease risk factors ■ risk factors ■ sleep deprivation

Heart disease and stroke remain the leading causes of morbidity and mortality in the United States and globally.¹ Research highlights insufficient sleep as a key risk factor for poor cardiovascular health, showing that sleep durations of ≤ 7 or ≥ 9 hours are linked to heart disease and other risk factors for cardiovascular disease (CVD).^{2–10} In 2022, this evidence prompted the American Heart Association to include getting healthy sleep into Life's Essential 8, recommending 7 to 9 hours of sleep per night for adults aged ≥ 20 years, based on self-report.² Sleep is a multidimensional concept, including efficiency, timing, daytime sleepiness, regularity, architecture, disordered sleep (including apnea), and perceived or self-reported sleep quality/satisfaction.^{11,12} However, due to insufficient evidence linking these other components to CVD and the practical challenges of measuring them in real-world settings, they were not included in Life's Essential 8.^{2,11,12}

Despite the recommendation that adults should get 7 to 9 hours of sleep per night,² many do not meet this target.^{11,12} Furthermore, a recent presidential advisory from the American Heart Association suggested that while poor diet, inadequate physical activity, and smoking are expected to improve, inadequate sleep duration

may continue to worsen through 2050.¹³ Long working hours, excessive stress, and advances in technological entertainment are aspects of modern society, which have contributed to worsening sleep.¹⁴ A better understanding of sleep-related cardiovascular risk has the potential to inform risk-reducing strategies. Thus, the purpose of this brief review is to summarize current evidence on the relationship of multidimensional sleep with heart disease and stroke, as well as the effects of sleep interventions on these conditions.

METHODS

We used PubMed's core clinical journals filter to search for articles published in high-impact, clinically useful journals.¹⁵ We used MeSH and search terms related to sleep, sleep quality, sleep duration, and the cardiovascular risk factors and events or diagnoses of interest. Original research studies were eligible for this review if they were published in English from 2020 to 2025, included adults (>18 years), assessed sleep through ≥ 1 of its concepts, and evaluated cardiovascular risk factors (overweight/obesity, dyslipidemia, hypertension, and type 2 diabetes [T2DM]/prediabetes) and events or diagnoses (atrial fibrillation [AF], stroke, myocardial infarction [MI], peripheral arterial disease [PAD], and atherosclerosis by imaging) as outcomes.

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Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
aHR	adjusted hazard ratio
AoU	All of Us
cIMT	carotid intima-media thickness
CVD	cardiovascular disease
HR	hazard ratio
MESA	Multi-Ethnic Study of Atherosclerosis
MI	myocardial infarction
MR	Mendelian randomization
PAD	peripheral arterial disease
T2DM	type 2 diabetes

Highlights

- This review highlights that most of the evidence in the past 5 years continues to focus on the relationship between sleep duration and cardiovascular disease risk, and the ongoing need to better understand how other sleep dimensions impact cardiovascular health.
- This review highlights emerging evidence that sleep regularity and architecture may be implicated in cardiovascular disease risk.
- This review emphasizes the need to move beyond relying solely on self-report and to use wearable devices for collecting long-term sleep data in large cohorts.

Studies that looked at the relationship between Life's Essential 8, solely as a composite score and not sleep individually, and the outcomes of interest were excluded.

First, we discuss how the various sleep dimensions are associated with CVD risk. A recent American Heart Association Scientific Statement on multidimensional sleep and cardiometabolic health served as the conceptual framework for the sleep dimensions in this review: regularity/rhythmicity, satisfaction/quality, alertness/sleepiness, timing, efficiency, duration, disturbed sleep, and sleep architecture.¹² We do not include the dimension of disturbed/disordered sleep in this review, as the relationship between sleep disorders, such as sleep apnea and insomnia, and sleep disordered breathing and CVD has been well documented.^{16–18} Second, we summarize and critique the sleep measurements utilized in the included studies. Finally, we discuss gaps in the evidence and directions for future research.

SLEEP-RELATED RISK OF HEART DISEASE AND STROKE

The Table summarizes key articles that examined how sleep duration, timing, regularity, sleepiness, and architecture relate to cardiovascular risk factors and conditions.

Cardiovascular Risk Factors

We reviewed studies that evaluated the relationship between sleep and the health factors of Life's Essential 8: overweight/obesity, dyslipidemia, hypertension, and T2DM.

Duration

As noted above, self-reported sleep duration is an established health behavior influencing CVD risk.² Recent studies using Mendelian randomization (MR), self-report, and wearable data have further explored this relationship.

The Fitbit data from the All of Us (AoU) Research Program (n=6785) demonstrated that higher average daily sleep duration was associated with reduced risk of obesity (hazard ratio [HR], 0.90 [95% CI, 0.83–0.98]).⁴¹ This study also revealed a J-shaped relationship between

sleep duration and hypertension, with participants averaging 5 (HR, 1.29 [95% CI, 1.09–1.54]) or 10 (HR, 1.61 [95% CI, 1.01–2.58]) hours of sleep at higher risk. An MR study among UK Biobank participants (n=404 044; 56.2±8.1 years) found that short sleep (≤6 hours) was associated with increased risk of hypertension (odds ratio, 1.15 [95% CI, 1.09–1.22]), but long sleep (≥9 hours) was not.²⁰ Another 2 MR studies assessing genetic variants associated with sleep duration found no causal relationship between sleep duration and the risk of T2DM or dyslipidemia.^{21,33}

These study findings align with prior evidence, showing that both self-reported and wearable-measured sleep duration are associated with the risk of obesity and hypertension. Long-sleep duration's association with cardiovascular risk factors may vary by population, or long-sleep duration may be a marker of poor general health.⁴⁴ The results from the MR studies were more mixed, indicating that findings may vary by methodology.

Timing

Two large MR studies in individuals of European ancestry found that genetic variants associated with self-reported chronotype or sleep timing (ie, being a morning versus evening person) had no causal relationship with dyslipidemia or T2DM.^{21,33} Most prior studies and reviews examining sleep timing suggest an association of later sleep onset with obesity and CVD; however, the definitions (eg, a bedtime later than midnight) used have been heterogeneous.¹²

Regularity

Sleep duration irregularity (SD of average daily sleep duration), assessed via a Fitbit in the AoU Research Program, was associated with a higher risk of obesity (HR, 1.21 [95% CI, 1.08–1.37]), hyperlipidemia (HR, 1.39 [95% CI, 1.20–1.61]), and hypertension (odds ratio, 1.56 [95% CI, 1.35–1.81]).⁴¹ Another prospective study among UK Biobank participants (n=73 630) leveraging an accelerometer found that moderately irregular

Table. Select Studies Evaluating the Association of Sleep with Cardiovascular Risk Factors and Diseases

Author, year	Design, sample size	Sleep dimensions (measurement)	Outcome associations		
			Positive	Negative	Null
Agudelo et al, 2023 ¹⁹	Cross-sectional, 1553	SD and DSIP (Self-report)	Carotid plaque presence (SD) and total plaque area (SD)		cIMT (SD and DSIP), carotid plaque presence (DSIP), and total plaque area (DSIP)
Ai et al, 2021 ²⁰	MR, 404 044	SD (GV)	MI (SD-short), hypertension (SD-short), and AF (SD-short)	Hypertension (SD-linear) and AF (SD-linear)	PAD, IS, HS, and TIA
Bos et al, 2021 ²¹	MR, 38 618	SD and ST (GV)			Concentration of very-large HDL-C particles (SD and ST), phospholipids in very-large HDL-C particles (SD and ST), and other metabolomics (SD and ST)
Chaput et al, 2024 ²²	Prospective, 73 630	SR (nCA)	Incident-T2DM		
Chaput et al, 2025 ²³	Prospective, 72 269	SR (nCA)	Incident-MI and incident-stroke		
Cheng et al, 2022 ²⁴	Prospective, 261 297	SD (self-report)	Incident-stroke and incident-hypertension		
Fang et al, 2024 ²⁵	Longitudinal, 9883	SD and SDR (Self-report)	Incident-dyslipidemia and incident-stroke		Incident-hypertension
Full et al, 2023 ²⁶	Longitudinal, 2032	SR (nCA)	CAC (SDR and STR), carotid plaque (SDR), and ABI (SDR)		cIMT (SDR and STR), carotid plaque (STR), and ABI (STR)
Guo et al, 2023 ²⁷	Cross-sectional, 240	SD (self-report)		IM-GSM	cIMT (Heart Association)
Han et al, 2023 ²⁸	Prospective, 18 876	SD (self-report)	Secondary: incident-IS and incident-PAD		
Han et al, 2024 ²⁹	Cross-sectional, 76 128	SD (self-report)	Secondary: stroke		
He et al, 2022 ³⁰	Prospective, 69 524	ST, DS, SD, DSIP, and CSS (self-report)		Secondary: incident-stroke (SD, DSIP, and CSS)	Secondary: incident-stroke (ST)
Huang et al, 2020 ³¹	Prospective, 1992	SR (nCA)	Incident-CVD		
Huang et al, 2025 ³²	Prospective, 86 219	SR (nCA)	Secondary: incident-stroke and incident-MI		
Jia et al, 2022 ³³	MR, IS: 440 328 T2DM: 898 148	DSIP, DN, ST, and SD (GV)	T2DM (DN)		T2DM (DSIP, ST, and SD) and IS (DSIP, DN, ST, and SD)
Kario et al, 2021 ³⁴	Prospective, 2253	SD (self-report)	Incident-stroke		
Kianersi et al, 2023 ³⁵	Prospective, 1259	SA (PSG)		Incident-T2DM	
Li et al, 2024 ³⁶	Prospective, 19 915	SD (self-report)		Secondary: incident-stroke	
Nambiema et al, 2023 ³⁷	Prospective, 11 347	CSS (self-report and PSG)		Incident-CVD (CSS) and incident-CHD (CSS)	Incident-stroke (CSS)
Souza et al, 2021 ³⁸	Cross-sectional, 2009	SD (nCA)			cIMT
Wang et al, 2023 ³⁹	Prospective, 29 211	ST, SD, DS, DSIP, and CSS (self-report)	Secondary: incident-stroke (CSS and SD)		Secondary: incident-stroke (ST and DSIP)
Yang et al, 2025 ⁴⁰	Prospective, 21 129	DSIP (self-report)	Secondary: incident-stroke and incident-MI		
Zheng et al, 2024 ⁴¹	Longitudinal, 6785	SD, SR, and SA (CA)	Incident-AF (SA-light), incident-obesity (SR), incident-hyperlipidemia (SR), and incident-hypertension (SD and SR)	Incident-AF (SA-deep) and incident-obesity (SD)	Incident-AF (SD and SR), incident-hypertension (SA), and incident-obesity (SA)
Zhou et al, 2020 ⁴²	Longitudinal, 31 750	SD, SDR, DN, and SQ (self-report)	Incident-stroke (SD, SDR, and DN) Secondary: incident-ischemic stroke (SD and DN)	Incident-stroke (SQ) Secondary: incident-ischemic stroke (SQ) and incident-hemorrhagic stroke (SQ)	Secondary: incident-hemorrhagic stroke (SD and DN)
Zhu et al, 2024 ⁴³	Prospective, 14 543	SD (self-report)			Incident-PAD

Unless otherwise noted, outcomes are primary. ABI indicates ankle-brachial index; AF, atrial fibrillation; BMI, body mass index; CA, commercial accelerometer; CAC, coronary artery calcium; CHD, coronary heart disease; cIMT, carotid intima-media thickness; CSS, composite sleep score of individual components; CVD, cardiovascular disease; DN, daytime napping; DS, disturbed sleep; DSIP, daytime sleepiness; GV, genetic variant; HDL-C, high-density lipoprotein-cholesterol; HS, hemorrhagic stroke; IM-GSM, gray-scale median of the intima-media complex; IS, ischemic stroke; MI, myocardial infarction; MR, Mendelian randomization; nCA, noncommercial accelerometer; PAD, peripheral arterial disease; PSG, polysomnography; SA, sleep architecture; SD, sleep duration; SDR, sleep duration regularity; SQ, sleep quality; SR, sleep regularity; ST, sleep timing; STR, sleep timing regularity; T2DM, type 2 diabetes; and TIA, transient ischemic attack.

(HR, 1.35 [95% CI, 1.19–1.53]) and irregular sleepers (HR, 1.38 [95% CI, 1.20–1.59]), based on variations in bedtime, wake-up time, and sleep interruptions, were at higher risk of developing T2DM.²² Regular sleep offsets the risk of developing T2DM among participants with abnormal sleep duration.²²

A study among middle-aged and older Chinese adults (n=9883) found that those who consistently self-reported short sleep (mean, 4.5–5.1 hours/night) over 4 years had an increased risk of developing dyslipidemia (HR, 1.22 [95% CI, 1.01–1.49]) compared with individuals with consistently higher sleep duration (mean, 6.8–7.3 hours/night) while adjusting for body mass index and other covariates.²⁵ Individuals with initially long sleep who began to sleep less, or initially short sleep, who began to sleep more, did not have a higher risk of developing dyslipidemia.²⁵

A prior review found that irregular sleep duration was likely associated with body mass index.⁴⁵ The findings from this review build upon prior work by demonstrating that objectively measured sleep irregularity is also associated with hyperlipidemia, hypertension, and T2DM.⁴¹

Daytime Sleepiness and Napping

MR of largely middle-aged and older adults from 32 European-ancestry studies (n=898 148) found that daytime napping (odds ratio, 1.56 [95% CI, 1.21–2.02]), but not sleepiness, was associated with a higher risk of T2DM.³³ This study builds on a prior systematic review finding that long daytime napping (>1 hour/d) is associated with higher odds of CVD risk factors, while, among older adults only, short daytime napping is also linked to increased CVD risk factor odds.⁴⁶

Architecture

The relationship between sleep architecture and the amount of time spent in different sleep stages was evaluated in 2 studies. Fitbit data from the AoU Research Program did not demonstrate associations between the average percent time spent in rapid eye movement, light, or deep sleep and obesity or hypertension.⁴¹ However, among participants in the MESA (Multi-Ethnic Study of Atherosclerosis) who underwent 1 night of polysomnography (n=1259), individuals who spent more time in deep sleep had a lower risk of T2DM (fourth-quartile HR, 0.32 [95% CI, 0.10–0.97]; third-quartile HR, 0.34 [95% CI, 0.15–0.77]; and second-quartile HR, 0.47 [95% CI, 0.26–0.87]).³⁵ These divergent findings may stem from differences in measurement techniques, with polysomnography being the gold standard for assessing sleep stages.

Cardiovascular Events and Diagnoses

We reviewed studies that evaluated the relationship between sleep and the following cardiovascular events

and diagnoses: AF, MI, stroke, PAD, and atherosclerosis by imaging.

Duration

Genetic variants linked to short-sleep duration were associated with risk of AF and MI (L-shaped relationship), suggesting causality.²⁰ However, genetic variants of short-sleep duration were not associated with stroke or coronary artery disease,^{20,33} and those of long-sleep duration were also not associated with AF, MI, stroke, or PAD.^{20,33} Neither short nor long-sleep duration, assessed via wearables (Fitbit and Actiwatch), was associated with incident-AF or atherosclerosis by carotid intima-media thickness (cIMT).^{38,41}

Self-reported short-sleep duration was associated with a higher risk of stroke across multiple studies^{25,28,29,34,39} but not all.^{24,42} In addition, a relationship was not established for PAD among patients with T2DM in 2 prospective studies^{28,43} or atherosclerosis (carotid plaque presence, total plaque area, or cIMT) in a cross-sectional study.¹⁹ No clear relationship was found between self-reported long-sleep duration and CVD, with some studies reporting positive relationships for stroke,^{24,28,39,42} PAD among patients with T2DM,²⁸ and atherosclerosis (carotid plaque presence and total plaque area),¹⁹ while other studies did not find an association (stroke,^{29,33,34} PAD among patients with T2DM,⁴³ and cIMT^{19,27}). Additional studies found that increased sleep duration or sleep duration in a normal range was associated with a lower risk of stroke.^{30,36,41}

One of these studies found that individuals with short-sleep duration, with nighttime systolic blood pressure >120 mm Hg, have a significantly higher risk of stroke (HR, 4.80 [95% CI, 1.57–14.65]) while controlling for antihypertensive medication use and office systolic blood pressure.³⁴ Another study found that while both short- and long-sleep durations were associated with a higher risk of stroke for individuals with T2DM (HR, 1.30 [95% CI, 1.12–1.51]), this was not the case for individuals with normal glucose tolerance or prediabetes.³⁹ Fang et al²⁵ found that individuals with consistently short-sleep duration and those with previous short-sleep duration that improved both had a higher risk of stroke. The studies that evaluated the relationship between sleep duration and atherosclerosis were cross-sectional and did not evaluate the relationship between sleep duration and MI, stroke, or PAD.^{19,27,38} This gap in the evidence makes it difficult to discern the pathway by which sleep duration may be associated with cardiovascular events.

Timing

Prior evidence has largely demonstrated a relationship between later sleep timing and worse CVD risk; however, the quality of evidence was low to moderate.⁴⁷ Two prospective cohort studies included in this review, among UK Biobank participants with hypertension and T2DM,

found no relationship between self-reported bedtime and stroke.^{30,39} These findings contrast prior research; additional high-quality studies are needed to clarify this relationship.

Regularity

The AoU Research Program did not find an association between sleep duration irregularity and AF as assessed via Fitbit over a median of 4.5 years of follow-up.⁴¹ However, a study among UK Biobank participants ($n=86219$) with 7 days of actigraphy from 2013 to 2015 found a higher risk of stroke (HR, 1.14 [95% CI, 1.03–1.27]) and MI (HR, 1.15 [95% CI, 1.04–1.27]) for every 1-hour increase in sleep duration SD.³² This association between sleep irregularity and stroke was more pronounced for ischemic (HR, 1.21 [95% CI, 1.08–1.37]) rather than hemorrhagic stroke, men (HR, 1.27 [95% CI, 1.11–1.44]), and those with an average sleep duration >8 hours (HR, 1.73 [95% CI, 1.32–2.27]).³² This association between sleep irregularity and MI was greater for individuals ≥ 60 years (HR, 1.29 [95% CI, 1.16–1.44]) and with a family history of CVD (HR, 1.33 [95% CI, 1.19–1.49]).³²

Another prospective study among UK Biobank participants ($n=72269$), leveraging an accelerometer to assess sleep, found that irregular sleepers were at higher risk of MI (HR, 1.23 [95% CI, 1.11–1.36]) and stroke (HR, 1.22 [95% CI, 1.01–1.48]).²³ Furthermore, they found that obtaining sufficient sleep duration offsets the risk for MI and stroke among moderately irregular but not irregular sleepers. Similarly, a prospective study among 1992 MESA participants found that greater irregularity in actigraphy-measured sleep duration (>120 minutes; HR, 2.14 [95% CI, 1.24–3.68]; $P_{\text{trend}}=0.002$) and sleep-onset timing (>90 minutes; HR, 2.11 [95% CI, 1.13–3.91]; $P_{\text{trend}}=0.002$) were associated with a higher risk of CVD.³¹

In MESA ($n=2032$), participants with greater irregular sleep duration (>120 minutes) were also more likely to have a higher coronary artery calcium burden (prevalence ratio, 1.33 [95% CI, 1.03–1.71]), carotid plaque (prevalence ratio, 1.12 [95% CI, 1.01–1.23]), and abnormal ankle-brachial index (prevalence ratio, 1.75 [95% CI, 1.03–2.95]) but not abnormal cIMT.²⁶ Participants in MESA with irregular sleep-onset timing (>90 minutes) were more likely to have a higher coronary artery calcium burden (prevalence ratio, 1.39 [95% CI, 1.07–1.82]) but not carotid plaque, abnormal ankle-brachial index, or abnormal cIMT.²⁶ Finally, Zhou et al⁴² found that participants with consistently long self-reported sleep duration (≥ 9 hours/night) and who transitioned from normal (7–9 hours/night) to long-sleep duration from 2008 to 2010 to 2013 had a higher incidence of stroke.

Daytime Sleepiness and Napping

Several studies found that frequent daytime napping, midday napping for longer than 90 minutes, or sleepiness

was associated with a higher risk of stroke,^{30,40,42} while others did not.^{33,39} Among 31750 retired employees from the Dongfeng-Tongji cohort, self-reported midday napping of >90 minutes was associated with increased risk of incident total (adjusted HR [aHR], 1.25 [95% CI, 1.03–1.53]) and ischemic (aHR, 1.27 [95% CI, 1.01–1.60]) stroke but not incident-hemorrhagic stroke.⁴² In addition, participants who reported both long-sleep duration (≥ 9 hours/night) and long midday napping (>90 minutes) had a higher risk of incident-stroke (aHR, 1.85 [95% CI, 1.28–2.66]).⁴² Among UK Biobank participants with T2DM ($n=21129$), self-reported daytime napping was associated with a higher risk of MI (HR, 1.44 [95% CI, 1.01–1.92]).⁴⁰ In a cross-sectional analysis, Agudelo et al¹⁹ did not find a relationship between daytime sleepiness and atherosclerosis, as assessed by carotid plaque presence, total plaque area, or cIMT. These findings align with a prior systematic review and meta-analysis, which found that daytime sleepiness was associated with stroke and coronary heart disease.⁴⁸ Only one of the studies evaluated the length of daytime napping.

Architecture

Using Fitbit data, the AoU Research Program found that a higher average percent time spent in deep sleep was associated with lower risk of developing AF (HR, 0.59 [95% CI, 0.35–0.99]).⁴¹ This relationship is thought to be related to the impact of sleep stages on signaling from the autonomic nervous system to the atria.

Sleep Quality

We identified 2 longitudinal studies, published within the last 5 years, which assessed the relationship between sleep quality and CVD, coronary heart disease, or stroke.^{37,42} The first consisted of 31750 Chinese retirees recruited between September 2008 and June 2010 and followed until 2013.⁴² Participants self-reported their sleep quality as good, fair, or poor/very poor with frequent use of hypnotics over the past 6 months. Compared with those with self-reported good sleep quality, participants with poor sleep quality at night had a higher risk of incident total (aHR, 1.29 [95% CI, 1.09–1.52]), ischemic (aHR, 1.28 [95% CI, 1.05–1.55]), and hemorrhagic (aHR, 1.56 [95% CI, 1.07–2.29]) stroke.⁴² Participants who reported either short (<7 hours/night) or long (≥ 9 hours/night) sleep duration, as well as poor sleep quality, had a higher risk of incident-stroke (short-sleep duration and poor sleep quality; aHR, 1.84 [95% CI, 1.25–2.70]; long-sleep duration and poor sleep quality; aHR, 1.82 [95% CI, 1.33–2.48]).⁴²

A study combining data across 2 cohort studies (the Paris Prospective Study III and the CoLausPsyCoLaus study) assessed sleep as a composite score via self-report (chronotype, sleep duration, insomnia symptoms, sleep apnea, and daytime sleepiness) and polysomnography (sleep duration, insomnia, and sleep apnea).³⁷ This study found that a higher (ie, better) self-reported

sleep score was associated with reduced risk of CVD and coronary heart disease but not stroke.³⁷ However, for this study, the number of stroke events was modest (ie, 72 cases).³⁷ Similar results were found using the polysomnography score.³⁷ A prior meta-analysis found that poor sleep quality was associated with a significant and moderate increase in the risk of coronary heart disease but not coronary heart disease mortality, CVD mortality, stroke, or CVD.⁴⁹ This meta-analysis included studies defining sleep quality widely.⁴⁹

SLEEP MEASUREMENT

Most studies used self-reported sleep data, while some directly measured sleep via wearable devices and polysomnography or genetic variants as proxy measures. The Figure depicts the sleep dimensions and their measurement tools, alongside cardiovascular risk factors and diseases evaluated in this review. Polysomnography has long been considered the gold standard for sleep measurement due to its ability to measure physiological changes and sleep architecture.⁵⁰ However, polysomnography is costly, typically only 1 night of data are collected, and it can result in decreased sleep efficiency, sleep latency, and rapid eye movement sleep due to an uncomfortable sleeping environment.⁵⁰ Thus, it is not feasible to leverage polysomnography for longitudinal sleep assessment.

Self-report can offer valuable insight into sleep satisfaction, quality, regular patterns, daytime sleepiness, and sleep disorder symptoms. However, when assessing sleep duration, regularity, timing, efficiency, and architecture, objective measures have advantages. Prior studies show that retrospective questionnaires overestimate sleep duration; Lauderdale et al⁶¹ demonstrated that this overestimation was more pronounced for short sleep (<7 hours).⁵² Nevertheless, Life's Essential 8 assesses sleep based on self-reported sleep duration, which has a large body of literature associated with it and practical advantages of not requiring a device.

Of the studies involving wearable devices, one used a Fitbit to collect data over a median of 4.5 years,⁴¹ while others used noncommercial accelerometers to collect data over 7 consecutive days.^{22,23,26,31,32,38} Compared with polysomnography, standard noncommercial accelerometers (without photoplethysmography) have been validated for total sleep time, sleep latency, and wake after sleep onset.⁵⁰ While they can reasonably be used to longitudinally assess sleep in a free-living environment over a fairly short period of time, they do tend to overestimate total sleep time and sleep efficiency in comparison to polysomnography, and their ability to capture wake detection could be improved.⁵⁰

In comparison to standard noncommercial accelerometers, commercial wearable devices with accelerometers and photoplethysmography (eg, Fitbit, Apple Watch, and

Garmin) may be more accurate in assessing sleep and wake detection.⁵⁰ This is because the photoplethysmography allows for heart rate measurement, which is generally highest during wake periods and rapid eye movement sleep, and lowest during deeper sleep.⁵⁰ The combined heart rate and accelerometer data allow for more accurate sleep staging than standard noncommercial accelerometers.⁵⁰ However, noncommercial accelerometers, such as ActiGraph LEAP, have recently started to incorporate photoplethysmography.

Commercial wearable devices, such as Fitbit, also allow for longer longitudinal data collection than standard noncommercial accelerometers, as they leverage cloud-based storage and APIs to store large data sets remotely without the need for physical data retrieval. This may allow for the testing of sleep regularity and architecture across periods of health and illness. However, it is unclear how software updates may affect the proprietary algorithms and data collected over time.⁵⁰

The analysis of genetically predicted sleep traits is a complementary approach that may help with causal inference. However, a challenge with this data modality and the related articles assessed in this review is that it is largely only available among individuals of European ancestry,⁵³ which may result in the development of genetic testing that is only applicable for certain groups, further accentuating disparities. Also, only a portion of the variability in sleep characteristics has genetic underpinnings, while much variability is a function of behaviors and environment.

CONCLUSIONS, GAPS IN THE EVIDENCE, AND FUTURE DIRECTIONS

The incorporation of healthy self-reported sleep duration into the American Heart Association Life's Essential 8 was an important step for recognizing sleep as a modifiable risk factor for CVD. In the past 5 years, the majority of new evidence among large samples has continued to focus on the relationship between sleep duration and CVD risk, likely due to the ease of measuring this dimension via self-report. Beyond sleep duration, we need to better understand how other sleep dimensions impact cardiovascular health. The findings from this review show emerging, but still limited, evidence that sleep regularity and architecture may be implicated in CVD risk.

This review covers the latest evidence, within the past 5 years, and offers several additional strengths. It examines the relationship between the various sleep dimensions, including sleep architecture, which was not included in prior reviews, and CVD risk.¹² Furthermore, the studies included in this review leveraged multiple approaches to sleep measurement, including those via wearable devices, as well as genetic variants associated with sleep traits. We also discuss the advantages and disadvantages

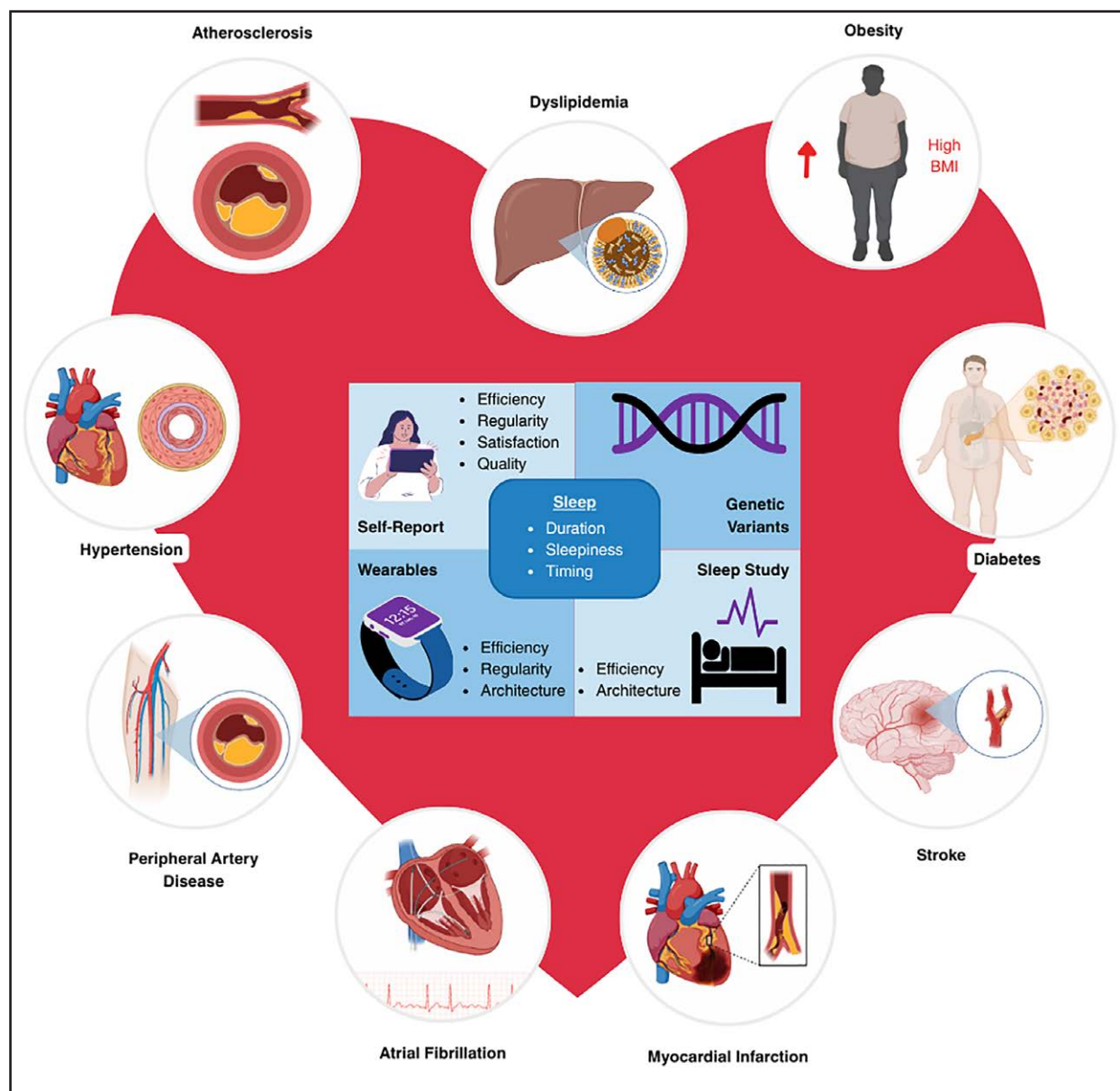


Figure. Measurement of sleep dimensions for cardiovascular risk ascertainment.

The diagram, framed within a heart, presents sleep dimensions and their measurement tools, alongside the cardiovascular risk factors and diseases evaluated in this review. The inner box highlights sleep dimensions that all measurement tools can assess. Canva was used to create this figure. Created in BioRender. BMI indicates body mass index.

of these more novel sleep measures. In addition, a broad range of cardiovascular risk factors and outcomes were examined, including atherosclerosis assessed by imaging. This review and the articles included also have limitations. First, none of the included studies assessed the relationship between sleep satisfaction or efficiency and CVD risk; only 2 studies assessed sleep quality; and only a few studies evaluated sleep timing. Second, the studies tended to include predominantly White participants of European ancestry, particularly in the genetic literature, limiting generalizability. Third, the studies that evaluated the relationship between sleep and atherosclerosis largely utilized cross-sectional designs and did

not evaluate other cardiovascular outcomes, making it challenging to discern the pathway by which sleep may be associated with cardiovascular events. Fourth, while we aimed to evaluate the effects of interventions related to sleep duration, timing, and regularity on CVD, we did not identify any recent, relevant studies.

Building on the findings and limitations of this review, we propose the following future research directions. First, given that sleep varies across countries and cultures,⁵⁴ future studies could examine the impact of cultural sleep differences on CVD risk. Second, the studies in this review evaluating sleep and PAD focused solely on sleep duration; therefore, future studies could

evaluate additional sleep dimensions in this condition. Third, additional research is needed on the impact of sleep timing on CVD risk. Fourth, studies are needed to determine whether atherosclerosis mediates the relationship between sleep and cardiovascular events, which is important for understanding the causal pathway by which sleep impacts cardiovascular outcomes.¹² Fifth, studies evaluating the impact of interventions striving to promote comprehensive sleep health for CVD prevention are needed.

Last, commercial wearable devices could be leveraged more frequently to objectively and longitudinally assess sleep duration, regularity, timing, efficiency, architecture, and circadian rhythms with cardiovascular health. Recent evidence has demonstrated that a suppressed 24-hour rest-activity rhythm is associated with a higher risk of stroke⁵⁵; future research could evaluate this relationship with other cardiovascular outcomes. Leveraging commercial wearables in these ways could provide key insights into how long-term sleep variability impacts cardiovascular health. The AoU Research Program study stands out for its use of a commercial wearable device for longitudinal data collection, but it included mostly White, middle-aged adults.⁴¹ Additional studies leveraging commercial wearable devices among older, racially/ethnically diverse individuals are needed. The adoption of commercial wearable devices is increasing (28% in 2019% to 36% in 2022), and national data from 2022 indicate that about 78% of adults would be willing to share their wearable device data with their clinicians.⁵⁶ A combination of self-report and long-term wearable sleep data may provide patients and clinicians with more actionable insights than self-report data alone.

Our understanding of sleep, as a multidimensional construct, is rapidly evolving as a potential risk factor for heart disease and stroke. Importantly, longitudinal, commercial wearable data are needed over longer periods to evaluate the relationship between long-term sleep and CVD risk. These data, alongside self-reports, could inform CVD risk-reduction strategies. Furthermore, additional evidence is needed on the effectiveness of structural and behavioral sleep interventions in improving CVD risk in modern society.

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